The gap series of an integer sequence: an arithmetic invariant governing subset-sum landscapes

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Abstract

For a finite sequence $A = (a_1, \dots, a_k)$ of positive integers we introduce the *gap series* $\Gamma(A) = \sum_{j=1}^{k} a_j 2^{-j} \in \mathbb{Q}$, an order-sensitive invariant equal to the generating function of A evaluated at $x = 1/2$. We show that Γ governs the combinatorics of subset sums of A.

Fix a target n, give the subsets S of A the energy $E(S) = |\sigma(S) - n|$, where σ is the sum, and take single-element flips as moves. Let $\mathrm{lm}_A(n)$ be the number of strict local minima and $\deg_A(n)$ the number of ground states. We prove a classification theorem for local minima; a window identity $W_D(A) = \Gamma(A) + (2D+1)2^{-k}$; and an exact stratification of $\mathrm{lm}_A(n)$ in which each stratum is a representation count r_{B_d} of a truncated subsequence. If those representation counts are *flat* to within a factor $1+\varepsilon$ on an explicit window, then

$$\frac{W_D(A)}{1+\varepsilon} \;\le\; \frac{\mathrm{lm}_A(n)}{\deg_A(n)} \;\le\; (1+\varepsilon)\, W_D(A).$$

This replaces the annealed (independence) approximation standard in the statistical-mechanics literature by a finite, checkable hypothesis.

We complement it with a lower bound. Writing $F_B(z) = \prod_{a \in B}(1 + z^a)$ for the generating polynomial of the representation counts, we show that $|F_B(i)| = 2^{|B|/2}$ exactly for *every* odd sequence B — the modulus-4 character never degenerates, whereas the modulus-6 one vanishes as soon as $3 \in B$. Consequently the representation counts are never equidistributed modulo 4: the spread of the residue-class totals is determined exactly by the parity of $|B|$, and the relative spread is always at least $2\sqrt{2} \cdot 2^{-|B|/2}$. We also determine which congruence is responsible: the geometric mean $M(q)$ of $|\cos(\pi r/q)|$ over r coprime to q equals $\frac{1}{2}|\Phi_q(-1)|^{1/\varphi(q)}$, and $\sup_{q \ge 3} M(q) = \sqrt{3}/2$ is attained at $q = 6$ alone. The primes, confined modulo 6 to the two residues realising this maximum, sit at rate $\sqrt{3}/2 \approx 0.866$ — above the universal floor $1/\sqrt{2} \approx 0.707$, so they are among the *least* flat odd sequences rather than the flattest.

We compute ε exactly for $k \le 32$: for the odd primes it decays geometrically at rate $\approx \sqrt{3}/2$, matching the modulus-6 prediction to within 1%, while for random odd sequences it decays strictly faster. Over 100 random sequences per $k \in \{8, 12, 16, 18, 20\}$ we find $(\mathrm{lm} / \deg)/\Gamma \to 1$ with standard deviation 0.0067 at $k = 20$ and $R^2 = 0.999$. A partial-correlation analysis shows that Γ subsumes the coefficient of variation of the weights, the statistic previously proposed for this role by Alyahya and Rowe.

All theorems below are formally verified in Lean 4 with Mathlib; no proof contains `sorry` and no additional axiom is used. The development is available at `https://github.com/mathlab0911/arithmetic-landscape`.

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Contents

1 Introduction

1.1 The gap series

Let $A = (a_1, \ldots, a_k)$ be a finite sequence of positive integers. Its *gap series* is

$$\Gamma(A) \;=\; \sum_{j=1}^{k} \frac{a_j}{2^j}.$$

Two elementary remarks fix the character of this quantity.

First, Γ is *order-sensitive*. It is not a function of the multiset $\{a_1, \ldots, a_k\}$: reordering the sequence changes its value. This distinguishes Γ from the statistics ordinarily attached to a set of weights — mean, variance, coefficient of variation — all of which are order-blind.

Second, writing $a_0 = 0$ and summing by parts,

$$\Gamma(A) \;=\; \sum_{j=0}^{k-1} \frac{a_{j+1} - a_j}{2^j} \;-\; \frac{a_k}{2^k},$$

so that for an increasing sequence Γ is a dyadically weighted sum of the *gaps* of A, dominated by the earliest ones. For the odd primes $3, 5, 7, 11, \ldots$ the small initial gaps — in particular the twin pairs $(3,5)$, $(5,7)$, $(11,13)$, $(17,19)$ — force Γ to be small: $\Gamma(3, 5, \ldots, 73) = 5609051/1048576 \approx 5.3492$, against ≈ 8.09 on average for a random odd sequence of the same length drawn from the same range.

The purpose of this paper is to show that this innocuous-looking quantity controls a genuinely combinatorial object: the local structure of the subset-sum problem for A.

Remark 1.1 (Terminology). The name is meant to record the second formula above: Γ is a dyadically weighted series in the gaps of A. It should not be confused with the *gap-sum sequence* of an integer sequence in the sense of Barry [Ba21], which is formed from the integers *absent* between consecutive terms; that construction and ours share no content beyond the word "gap". Readers who prefer an unambiguous name may read Γ as the *dyadic gap series*, or simply as the generating function of A evaluated at $x = 1/2$.

1.2 Subset-sum landscapes

Fix a target $n \in \mathbb{N}$. For $S \subseteq \{1, \ldots, k\}$ put $\sigma(S) = \sum_{i \in S} a_i$ and $E(S) = |\sigma(S) - n|$. Two configurations are adjacent if they differ in a single index. This is the standard 1-bit-flip landscape of the number partitioning problem (NPP); the case $n = \sigma(A)/2$ is the balanced partition problem. We study

$$\deg_A(n) = \#\{S : E(S) = 0\}, \qquad \mathrm{lm}_A(n) = \#\{S : E(S) < E(S') \text{ for all neighbours } S'\}.$$

Here lm counts the *valleys* of the landscape and is the standard measure of its ruggedness.

Our main results are the following; full statements appear in the sections indicated.

Theorem A (Classification; §3). Suppose all $a_i > 0$ and let $d = E(S)$. If $\sigma(S) \geq n$, then S is a strict local minimum if and only if $a_i > 2d$ for every $i \in S$. If $\sigma(S) \leq n$, then S is a strict local minimum if and only if $a_i > 2d$ for every $i \notin S$.

Thus membership in the set of local minima is decided by a condition involving only the elements of size at most $2d$; everything larger is irrelevant. This is the engine of the whole paper.

Theorem B (Window identity; §5). Let A be non-decreasing with all a_i odd and $a_i \leq 2D + 1$. Put $N_A(d) = \#\{i : a_i \leq 2d\}$ and $W_D(A) = 1 + 2\sum_{d=1}^{D} 2^{-N_A(d)}$. Then

$$W_D(A) \;=\; \Gamma(A) + \frac{2D+1}{2^k}.$$

The quantity W_D is the total measure of the "windows" in which a configuration at error d can be a local minimum, computed as if the elements were placed independently. Theorem B says that this measure is exactly the gap series, up to a tail term negligible for the sequences of interest ($73/2^{20} \approx 7 \times 10^{-5}$ for the first twenty odd primes).

Theorem C (Exact stratification; §6). Without any approximation,

$$\mathrm{lm}_A(n) \;=\; \deg_A(n) + \sum_{d\geq 1}\bigl(\mathrm{over}_A(n,d) + \mathrm{under}_A(n,d)\bigr),$$

where the strata vanish once $2d$ exceeds $\max_i a_i$, and each stratum equals a representation count of a truncated subsequence,

$$\mathrm{over}_A(n,d) = r_{B_d}(n+d), \qquad \mathrm{under}_A(n,d) = r_{B_d}(n-d-\sigma(I_d)),$$

with $I_d = \{i : a_i \leq 2d\}$ and $B_d = A|_{I_d^c}$. Moreover $\deg_A(n) = \sum_{J\subseteq I_d} r_{B_d}(n-\sigma(J))$.

Theorem D (Sandwich; §7). Suppose that for every $d \leq D$ the representation counts r_{B_d} vary by at most a factor $1+\varepsilon$ across the window $\{n-\sigma(J) : J \subseteq I_d\} \cup \{n+d,\ n-d-\sigma(I_d)\}$. Then

$$\frac{W_D(A)}{1+\varepsilon} \;\leq\; \frac{\mathrm{lm}_A(n)}{\deg_A(n)} \;\leq\; (1+\varepsilon)\,W_D(A).$$

Combining Theorems B and D, $\mathrm{lm}\,/\deg$ is pinned to $\Gamma(A)$ with a multiplicative error controlled by a single, explicitly computable number ε.

The flatness ε cannot be made to vanish arbitrarily fast. Write $F_B(z) = \prod_{a\in B}(1+z^a)$, so the representation counts r_B are the coefficients of F_B, and let $R_c = \sum_{m\equiv c\,(4)} r_B(m)$.

Theorem E (Universal obstruction; §7.3). For every finite sequence B of odd positive integers, $|F_B(i)| = 2^{|B|/2}$ exactly, whatever the residues of the elements. Consequently the counts R_c are never all equal; their spread is determined by the parity of $|B|$,

$$\max_c R_c - \min_c R_c = 2^{|B|/2} \;\; (|B| \text{ even}), \qquad 2^{(|B|-1)/2} \;\; (|B| \text{ odd}),$$

so the relative spread is always at least $2\sqrt{2}\cdot 2^{-|B|/2}$.

This is a lower bound, in contrast with Theorems B–D, and it is elementary: $1+i^a = 1\pm i$ has modulus $\sqrt{2}$ for *every* odd a, so the modulus-4 character never degenerates. The modulus-6 character does: if $3 \in B$ then $F_B(\zeta_6) = 0$.

The obstruction of Theorem E is a statement about the distribution of r_B across residue classes modulo 4, not about the pointwise flatness ε_d of Theorem D; we do not derive a lower bound for the latter, and §7.3 explains why the transfer needs an input we do not have. Numerically, however, the two sit at the same scale for random odd sequences, whereas for the primes ε_2 exceeds the modulus-4 scale by two orders of magnitude at $k = 32$: their obstruction decays at rate $\sqrt{3}/2 \approx 0.866$, against $1/\sqrt{2} \approx 0.707$. In that sense the primes are not the flattest odd sequences but among the least flat.

We emphasise what Theorem D does *not* assume. The statistical-mechanics treatment of the NPP proceeds through the annealed approximation, in which configurations are treated as independent. Theorems C and D dispense with this: the decomposition is an identity, and the only input is the flatness of an explicit arithmetic function on an explicit finite window. Flatness is not an idealisation — it is a quantity one can compute for any given A, and we do so in §7.4.

1.3 Relation to previous work

The NPP has been studied intensively from the statistical-mechanics side. Ferreira and Fontanari [FF98] analysed the problem for i.i.d. weights uniform on $[0,1]$, identifying it with the ground-state problem for an infinite-range antiferromagnetic Ising model, and computed within the annealed approximation the fraction of metastable states — configurations stable under all single spin flips — showing that it vanishes like $N^{-3/2}$. Mertens [Me01] surveys the physicist's approach; Stadler, Hordijk and Fontanari [SHF03] study landscape statistics across the phase transition; and Borgs, Chayes and Pittel [BCP01] give a rigorous treatment of the transition and its finite-size scaling.

We stress that *the order of magnitude* $2^k k^{-3/2}$ *for the number of local minima of a random instance is not new here.* It is exactly the content of the annealed computation of [FF98]. Our contribution lies elsewhere, in two directions.

Deterministic sequences. The literature treats the weights as random and asks for typical behaviour. We fix a deterministic sequence — for instance the primes — and ask which of its *arithmetic* features control the landscape. The answer, Theorems B–D, is the gap series, an invariant with no probabilistic content.

Exactness. The annealed approximation is replaced by an identity plus a checkable hypothesis. To our knowledge Theorem C, and the reduction of lm / deg to the flatness of truncated representation counts, are new.

The closest precursor is Alyahya and Rowe [AR14], who observed empirically that the number of local optima in the 1-bit-flip NPP landscape is strongly *negatively* correlated with the coefficient of variation CV of the weights, and proposed an estimator based on CV. Their observation is correct and we reproduce it (§9). But CV is order-blind while Γ is not, and Theorem B shows it is Γ, not CV, that equals the window measure W_D. Partial-correlation analysis on 100 instances per size confirms the theoretical expectation: conditioning on CV leaves the Γ-correlation essentially intact, while conditioning on Γ removes almost all of the CV-correlation. In this precise sense Γ subsumes CV.

1.4 Formal verification

Every theorem stated in §§2–7 has been formalised in Lean 4 against Mathlib. Each formal statement is accompanied in the text by the name of the corresponding Lean declaration, and Appendix A lists the files, the declarations and the reproduction procedure. We have verified with `#print axioms` that no declaration depends on `sorryAx` or on any axiom beyond Lean's standard `propext`, `Classical.choice` and `Quot.sound`.

Two remarks on what this did and did not buy us. It did not merely confirm proofs already written on paper: formalisation of Theorem D revealed that the exponent could be improved from $(1+\varepsilon)^{\pm 2}$ to $(1+\varepsilon)^{\pm 1}$, since numerator and denominator may be controlled by the same flatness estimate. On the other hand, formalisation forced us to state hypotheses that are easy to leave implicit — most notably the non-decreasing hypothesis in Theorem B, whose necessity we exhibit by a two-element counterexample in §5.3.

2 Definitions, and the bridge to computation

2.1 The landscape

Throughout, A denotes a finite sequence of positive integers, presented either as a list $(a_1, \dots, a_k)$ or as a function $A : \mathbb{F}_k \to \mathbb{N}$ on the index type $\mathbb{F}_k = \{0, \dots, k-1\}$; §2.2 shows the two presentations agree. We write $\sigma(S) = \sum_{i \in S} a_i$ for $S \subseteq \mathbb{F}_k$, and $T = \sigma(A)$. For a target n,

$$E_n(S) = |\sigma(S) - n|, \qquad \mathrm{flip}(S, i) = \begin{cases} S \setminus \{i\} & i \in S, \\ S \cup \{i\} & i \notin S, \end{cases}$$

and S is a *strict local minimum* if $E_n(S) < E_n(\mathrm{flip}(S, i))$ for all i. We set

$$\mathrm{gs}_A(n) = \min_S E_n(S), \quad \deg_A(n) = \#\{S : \sigma(S) = n\}, \quad \mathrm{lm}_A(n) = \#\{S : S \text{ strict local min}\}.$$

(Lean: `ndist`, `subsetSums`, `gs`, `energy`, `flip`, `IsStrictLocalMin`, `lm`, `degCount`.)

2.2 The bridge

Implementations enumerate subsets by bitmasks over a list; proofs are more convenient over $\mathrm{Finset}(\mathbb{F}_k)$. That these describe the same object is not automatic, and we record it.

Proposition 2.1. *For $A : \mathbb{F}_k \to \mathbb{N}$ and $s \in \mathbb{N}$, $s \in \mathrm{subsetSums}(\mathrm{ofFn}\, A)$ if and only if there exists $S \subseteq \mathbb{F}_k$ with $\sigma(S) = s$.* (Lean: `mem_subsetSums_ofFn_iff`.)

Proposition 2.2. *$\mathrm{gs}_A(n)$ equals $\min_S E_n(S)$, and the minimum is attained.* (Lean: `gs_-ofFn_eq_energy_min`.)

Proposition 2.3 (Representability)**.** *$\mathrm{gs}_A(n) = 0$ if and only if n is a sum of a sub-sequence of A.* (Lean: `gs_eq_zero_iff`.)

Proposition 2.3 is the translation used throughout: the statement "the landscape has ground energy zero" is the statement "n is representable", and $\deg_A(n)$ is the number of representations, denoted $r_A(n)$ below.

3 The classification theorem

Theorem 3.1 (Classification)**.** *Let $a_i > 0$ for all i, let $S \subseteq \mathbb{F}_k$ and $d = E_n(S)$.*

1. *If $\sigma(S) \geq n$: S is a strict local minimum $\iff$ $a_i > 2d$ for all $i \in S$.*
2. *If $\sigma(S) \leq n$: S is a strict local minimum $\iff$ $a_i > 2d$ for all $i \notin S$.*

(Lean: `isStrictLocalMin_iff_of_ge`, `isStrictLocalMin_iff_of_le`, combined form `is-StrictLocalMin_iff_classCond`.)

Proof. Suppose $\sigma(S) = n + d$ with $d \geq 0$. Removing $i \in S$ changes the sum to $n + d - a_i$, whose error is $|d - a_i|$; this is $\geq d$ precisely when $a_i \geq 2d$, and $> d$ precisely when $a_i > 2d$. Adding $i \notin S$ only increases the error, since $a_i > 0$. The deficient case is symmetric. □

We claim no originality for Theorem 3.1: the computation is immediate, and some form of it is presumably known to anyone who has looked at the 1-bit-flip landscape of the NPP, though we have not located an explicit statement in the literature. We record it here because everything below is built on it, and because the formal development requires a precise version. What we do claim is the use made of it in §§5–7.

Corollary 3.2. *A ground state ($d = 0$) is always a strict local minimum.* (Lean: `isStrict-LocalMin_of_exact`.)

Corollary 3.3. *$\mathrm{lm}_A(n)$ equals the number of S satisfying the arithmetic condition of Theorem 3.1; in particular lm is a counting function of the sequence and target alone.* (Lean: `lm_eq_lmClassCount`.)

The force of Theorem 3.1 is a *locality* statement: whether S is a valley depends only on the elements of A that are at most $2d$, where d is the error at S. Since a landscape has few configurations at large error, and the constraint at error d involves $N_A(d) = \#\{i : a_i \leq 2d\}$ elements, the count of valleys is governed by how quickly $N_A(d)$ grows — that is, by the early gaps of A. Sections 5–7 make this precise.

4 Symmetry and parity

Throughout this section $T = \sigma(A)$.

Theorem 4.1 (Reflection). *For $n \le T$: $\mathrm{gs}_A(T-n) = \mathrm{gs}_A(n)$ and $\mathrm{lm}_A(T-n) = \mathrm{lm}_A(n)$.* (Lean: `gs_compl`, `lm_compl`.)

Proof. The map $S \mapsto S^c$ is an involution of the configuration space satisfying $\sigma(S^c) = T - \sigma(S)$, hence $E_{T-n}(S^c) = E_n(S)$; and it commutes with single-element flips, $\mathrm{flip}(S,i)^c = \mathrm{flip}(S^c,i)$. So it carries strict local minima for target n bijectively onto those for target $T-n$. $\square$

Theorem 4.2 (No flat edges). *If every a_i is odd, then $E_n(S) \neq E_n(\mathrm{flip}(S,i))$ for all S and i.* (Lean: `no_flat_edge`.)

Proof. $|x-n| \equiv x+n \pmod 2$, and flipping i changes $\sigma(S)$ by the odd number a_i, hence changes the parity of the error. $\square$

Corollary 4.3. *For odd A, strict local minima and weak local minima coincide.* (Lean: `isStrictLocalMin_iff_le_of_odd`.)

Corollary 4.3 is the reason we restrict to odd sequences: for such A — and the odd primes are the motivating example — the quantity $\mathrm{lm}_A(n)$ does not depend on which of the two standard definitions of "local minimum" one adopts. For sequences containing even elements the landscape has plateaus and the two counts genuinely differ.

5 The window identity

5.1 Windows

By Theorem 3.1, a configuration at error d is a local minimum precisely when the $N_A(d)$ elements of size $\le 2d$ all lie on a prescribed side. If the elements were placed independently and uniformly, the proportion of configurations satisfying this would be $2^{-N_A(d)}$. Summing over d, with the excess and deficient sides contributing separately and the ground level ($d = 0$) once, gives the *window series*

$$W_D(A) \;=\; 1 + 2\sum_{d=1}^{D} 2^{-N_A(d)}, \qquad N_A(d) = \#\{i : a_i \le 2d\}.$$

(Lean: `winCount`, `windowSum`, `windowSeries`; the $\mathbb{F}_k$ version is `winCountFin`, `winSeriesFin`.)

W_D is the annealed prediction for $\mathrm{lm}_A(n)/\deg_A(n)$. Sections 6–7 show in what exact sense it is correct; the present section evaluates it.

5.2 The identity

Theorem 5.1. *Let A be a non-decreasing sequence of odd positive integers with $a_i \le 2D+1$ for all i. Then*

$$W_D(A) \;=\; \Gamma(A) + \frac{2D+1}{2^k}.$$

(Lean: `windowSeries_eq_gapSeries`.)

Proof. Induction on the length of A, the inductive step resting on the pointwise identity

$$2^{-N_{a:t}(d)} \;=\; \tfrac{1}{2}\Big(2^{-N_t(d)} + \mathbf{1}[\,2d < a\,]\Big),$$

valid when $a \le b$ for all $b \in t$ (Lean: `window_term_cons`). Indeed if $a \le 2d$ then $N_{a:t}(d) = N_t(d) + 1$ and the indicator vanishes; if $2d < a$ then no element of $a : t$ is $\le 2d$, so both sides equal 1. Summing over $d \in [1, D]$ and using $\#\{d \le D : 2d < a\} = \lfloor a/2 \rfloor$ (here $a \le 2D+1$ is used) gives

$$\sum_{d=1}^{D} 2^{-N_{a:t}(d)} = \tfrac{1}{2}\Big(\sum_{d=1}^{D} 2^{-N_t(d)} + \lfloor a/2 \rfloor\Big),$$

whence $W_D(a : t) = 1 + \lfloor a/2 \rfloor + \frac{1}{2}(W_D(t) - 1)$. Substituting the inductive hypothesis $W_D(t) = \Gamma(t) + (2D+1)2^{-|t|}$ and using $\lfloor a/2 \rfloor = (a-1)/2$ for odd a, together with $\Gamma(a : t) = \frac{a}{2} + \frac{\Gamma(t)}{2}$, gives the claim. The base case $A = \varnothing$ reads $1 + 2D = 0 + (2D+1)$. □

Note that D may be taken arbitrarily large: increasing D adds terms 2^{-k} on the left, absorbed exactly by the tail term on the right. The natural choice for an increasing sequence is $D = (a_k - 1)/2$, the smallest value for which all elements are $\le 2D+1$; then $W(A) = \Gamma(A) + a_k 2^{-k}$, and for the first twenty odd primes the correction is $73/2^{20} \approx 6.96 \times 10^{-5}$.

5.3 Order matters

W_D depends only on the multiset underlying A, whereas Γ depends on the order. Theorem 5.1 can therefore hold only for one ordering, and it is worth exhibiting the failure explicitly.

Example 5.2. Take $A = (5, 3)$ and $D = 2$. Then $N_A(1) = 0$ and $N_A(2) = 1$, so $W_2(A) = 1 + 2(1 + \frac{1}{2}) = 4$, while $\Gamma(A) = \frac{5}{2} + \frac{3}{4} = \frac{13}{4}$ and $\Gamma(A) + \frac{5}{4} = \frac{9}{2} \neq 4$. Reordering to $(3, 5)$ restores the identity: $\Gamma = \frac{3}{2} + \frac{5}{4} = \frac{11}{4}$ and $\frac{11}{4} + \frac{5}{4} = 4$. Both computations are `#eval`-checked in the Lean development.

We return to this in §9: it is precisely the order-sensitivity of Γ that separates it from order-blind statistics such as the coefficient of variation, and Example 5.2 is the shortest demonstration that no order-blind statistic can play the role of Γ in Theorem 5.1.

6 Exact stratification

6.1 Strata

Fix A with $a_i > 0$ and a target n. Following Theorem 3.1, define for $d \ge 0$

$$\begin{aligned} \mathrm{over}_A(n, d) &= \#\{S : \sigma(S) = n + d,\ \forall i \in S,\ a_i > 2d\}, \\ \mathrm{under}_A(n, d) &= \#\{S : \sigma(S) + d = n,\ \forall i \notin S,\ a_i > 2d\}. \end{aligned}$$

(Lean: `overCount`, `underCount`.)

Lemma 6.1. *The set of strict local minima with $\sigma(S) = s$ has cardinality $\mathrm{over}_A(n, s-n)$ if $s \ge n$, and $\mathrm{under}_A(n, n-s)$ if $s < n$.* (Lean: `fiber_over`, `fiber_under`.)

Lemma 6.2. $\mathrm{over}_A(n, 0) = \mathrm{under}_A(n, 0) = \deg_A(n)$. *If $a_i \le 2d$ for all i and $d \ge 1$, then* $\mathrm{over}_A(n, d) = \mathrm{under}_A(n, d) = 0$. (Lean: `overCount_zero_eq_degCount`, `underCount_zero_eq_degCount`, `overCount_eq_zero`, `underCount_eq_zero`.)

Theorem 6.3 (Exact stratification)**.** *For $n \le T$ and any D with $a_i \le 2D+1$ for all i,*

$$\mathrm{lm}_A(n) \;=\; \deg_A(n) \;+\; \sum_{d=1}^{D} \big(\mathrm{over}_A(n, d) + \mathrm{under}_A(n, d)\big).$$

(Lean: `lm_eq_sum_strata` in the sum-indexed form; `lm_eq_lmPartial` and `lm_eq_lmPartial_of_le` in the d-indexed form used here.)

Proof. Fibring the set of strict local minima over the value $\sigma(S) \in [0, T]$ and applying Lemma 6.1 gives the sum-indexed identity. Re-indexing the excess part by $d = s - n$ and the deficient part by $d = n - s$ (the latter via reflection of a range sum), then extracting the $d = 0$ term and truncating both sums at D using Lemma 6.2, gives the stated form. □

We stress that Theorem 6.3 is an identity between integers. No independence assumption enters.

6.2 Strata are representation counts

Write $I_d = \{i : a_i \le 2d\}$, $N(d) = |I_d|$, $\sigma_d = \sigma(I_d)$, and let B_d be the subsequence of A indexed by I_d^c. As in §2.2, $r_B(m)$ denotes the number of subsets of B summing to m.

Theorem 6.4 (Fibres)**.** *Let $J \subseteq I$. Then*

$$\#\{S : \sigma(S) = n,\ S \cap I = J\} \ = \ r_{A|_{I^c}}(n - \sigma(J)).$$

(Lean: `fiber_card_eq_repCount`; the bijection is $S \mapsto S \setminus I$, with inverse $U \mapsto U \cup J$.)

Corollary 6.5. $\deg_A(n) = \sum_{J \subseteq I} r_{A|_{I^c}}(n - \sigma(J))$ *for any I.* (Lean: `degCount_eq_sum_-repCount`, combining Theorem 6.4 with `degCount_fiberwise`.)

Corollary 6.6. $\mathrm{over}_A(n, d) = r_{B_d}(n + d)$ *and, for $d \le n$,* $\mathrm{under}_A(n, d) = r_{B_d}(n - d - \sigma_d)$. (Lean: `overCount_eq_repCount`, `underCount_eq_repCount`; these are Theorem 6.4 with $J = \varnothing$ and $J = I_d$ respectively.)

6.3 What has been achieved

Combining the above,

$$\frac{\mathrm{lm}_A(n)}{\deg_A(n)} \ = \ 1 + \sum_{d=1}^{D} \frac{r_{B_d}(n + d) + r_{B_d}(n - d - \sigma_d)}{\sum_{J \subseteq I_d} r_{B_d}(n - \sigma(J))}.$$

Every quantity on the right is a representation count of a truncated subsequence of A. The ratio of valleys to ground states has been expressed *exactly* in terms of the arithmetic function r, with no probabilistic input whatever.

The denominator is a sum of $2^{N(d)}$ values of r_{B_d} at the points $n - \sigma(J)$, $J \subseteq I_d$; the numerator consists of two further values of the same function. If r_{B_d} were constant on the window spanned by these $2^{N(d)} + 2$ points, the d-th summand would equal exactly $2 \cdot 2^{-N(d)}$ and the total would be exactly $W_D(A) = \Gamma(A) + (2D + 1)2^{-k}$ by Theorem 5.1.

The annealed approximation of the physics literature is thus revealed to be exactly the assertion that r_{B_d} is constant on this window. It is not; but it is nearly so, and "nearly" is quantifiable. That is the content of §7.

7 Flatness and the sandwich

7.1 The abstract lemma

Lemma 7.1. *Let s be a finite non-empty index set, $g : s \to \mathbb{Q}_{>0}$, $v \in \mathbb{Q}_{>0}$ and $\varepsilon \ge 0$. If $g(i) \le (1 + \varepsilon)v$ and $v \le (1 + \varepsilon)g(i)$ for all $i \in s$, then*

$$\frac{1}{(1 + \varepsilon)\,|s|} \ \le \ \frac{v}{\sum_{i \in s} g(i)} \ \le \ \frac{1 + \varepsilon}{|s|}.$$

(Lean: `ratio_bounds_of_flat`.)

Proof. $\sum g \geq |s|\, v/(1+\varepsilon)$ gives the upper bound; $\sum g \leq |s|(1+\varepsilon)v$ gives the lower. □

Elementary as it is, Lemma 7.1 is the whole mechanism: it converts flatness of a family into a bound on the share of any one member.

7.2 The sandwich

Definition 7.2. Say A is *ε-flat at (n,d)* if $r_{B_d}(m') \leq (1+\varepsilon) r_{B_d}(m)$ for all $m, m' \in \mathrm{Win}_d$, where

$$\mathrm{Win}_d = \{n - \sigma(J) : J \subseteq I_d\} \cup \{n+d,\ n-d-\sigma_d\}.$$

Theorem 7.3 (Sandwich)**.** *Let $a_i > 0$ be odd, $n \leq T$, and let D satisfy $a_i \leq 2D+1$ for all i. Suppose $\deg_A(n) > 0$ and A is ε-flat at (n,d) for every $1 \leq d \leq D$. Then*

$$\frac{W_D(A)}{1+\varepsilon} \;\leq\; \frac{\mathrm{lm}_A(n)}{\deg_A(n)} \;\leq\; (1+\varepsilon)\, W_D(A),$$

and by Theorem 5.1 the outer terms equal $\left(\Gamma(A) + (2D+1)2^{-k}\right)(1+\varepsilon)^{\mp 1}$. (Lean: `stratum_-ratio_bounds`, `sandwich_partial`, `sandwich_total`.)

Proof. Fix d. By Corollary 6.5 the denominator $\deg_A(n)$ is the sum of the $2^{N(d)}$ numbers $r_{B_d}(n - \sigma(J))$, all of which lie within a factor $1+\varepsilon$ of $\mathrm{over}_A(n,d) = r_{B_d}(n+d)$ by ε-flatness. Lemma 7.1 gives

$$\frac{2^{-N(d)}}{1+\varepsilon} \;\leq\; \frac{\mathrm{over}_A(n,d)}{\deg_A(n)} \;\leq\; (1+\varepsilon)\, 2^{-N(d)},$$

and the same for under. Summing over $1 \leq d \leq D$ and adding the $d=0$ term $\deg / \deg = 1$, then invoking Theorem 6.3, gives the result; the elementary inequalities $1 \leq 1+\varepsilon$ and $(1+\varepsilon)^{-1} \leq 1$ absorb the ground term. □

Remark 7.4. The exponent is $(1+\varepsilon)^{\pm 1}$, not $(1+\varepsilon)^{\pm 2}$ as a first analysis suggests, because the same flatness hypothesis controls numerator and denominator simultaneously. We record this because it was found only in the course of formalising the argument.

Remark 7.5. Theorem 7.3 is a finite statement over $\mathbb{Q}$. For any explicit A, n, D and ε all quantities are computable, so the theorem yields certified error bounds for individual instances rather than an asymptotic statement. The asymptotic question — whether ε may be taken to 0 as $k \to \infty$ — is the subject of §7.4 and §10.

7.3 A universal obstruction to flatness

Before measuring ε we record a lower bound for it, valid for every odd sequence and proved by an entirely elementary character computation. Recall $F_B(z) = \prod_{a \in B}(1 + z^a)$ and set $b = |B|$ and $R_c = \sum_{m \equiv c\,(4)} r_B(m)$ for $c \in \mathbb{Z}/4$.

Theorem 7.6 (Universal modulus-4 obstruction)**.** *Let B be any finite sequence of odd positive integers, $b = |B|$. Then*

1. *$|F_B(i)| = 2^{b/2}$, exactly, independently of the residues of the elements of B; in particular $F_B(i) \neq 0$. More precisely $F_B(i) = 2^{b/2} e^{i\pi(p-q)/4}$ where $p = \#\{a \in B : a \equiv 1\,(4)\}$ and $q = \#\{a \in B : a \equiv 3\,(4)\}$.*

2. *The spread of the residue-class totals is given* exactly *by the parity of b:*

$$\max_c R_c - \min_c R_c = \begin{cases} 2^{b/2} & b \text{ even}, \\ 2^{(b-1)/2} & b \text{ odd}. \end{cases}$$

3. *Since* $\sum_c R_c = 2^b$*, the relative spread equals* $4 \cdot 2^{-b/2}$ *for* b *even and* $2\sqrt{2} \cdot 2^{-b/2}$ *for* b *odd; in either case*
$$\frac{\max_c R_c - \min_c R_c}{\frac{1}{4}\sum_c R_c} \;\geq\; 2\sqrt{2} \cdot 2^{-b/2}.$$

(Lean: `normSq_one_add_I_pow_odd`, `max_abs_cos_sin_sq_ge`, `spread_lower_bound` give (i) and the inequality in (iii); `max_abs_cos_sin_sq`, `max_abs_cos_sin_quarter_even` and `max_abs_cos_sin_quarter_odd` give the parity dichotomy in (ii), the spread being $2^{b/2}\max(|\cos\varphi|, |\sin\varphi|)$ with $\varphi = \pi(p-q)/4$ and $p - q \equiv b \pmod 2$.)

Proof. (i) For odd a we have $i^a = \pm i$, so $1 + i^a$ is $1+i$ or $1-i$ according as $a \equiv 1$ or $3 \pmod 4$; both have modulus $\sqrt{2}$, and $1 \pm i = \sqrt{2}\, e^{\pm i\pi/4}$. Multiplying over B gives $F_B(i) = 2^{b/2} e^{i\pi(p-q)/4}$.

(ii) By discrete Fourier inversion on $\mathbb{Z}/4$,
$$R_c = \tfrac{1}{4}\sum_{j=0}^{3} i^{-jc} F_B(i^j) = \tfrac{1}{4}\Big(2^b + 2\,\mathrm{Re}\big(i^{-c}F_B(i)\big)\Big),$$
because $F_B(1) = 2^b$ and $F_B(-1) = \prod_{a\in B}(1 + (-1)^a) = 0$, every a being odd. Writing $\varphi = \pi(p-q)/4$, the quantity $\mathrm{Re}(i^{-c}F_B(i))/2^{b/2}$ runs over $\{\cos\varphi, \sin\varphi, -\cos\varphi, -\sin\varphi\}$ as c runs over $\mathbb{Z}/4$, so the spread is $2^{b/2} \cdot 2\max(|\cos\varphi|, |\sin\varphi|)/2$. By (i) φ is a multiple of $\pi/4$, and $p + q = b$ forces $p - q \equiv b \pmod 2$. If b is even then φ is a multiple of $\pi/2$ and $\max(|\cos\varphi|, |\sin\varphi|) = 1$; if b is odd then φ is an odd multiple of $\pi/4$ and the maximum is $1/\sqrt{2}$. This gives the two cases.

(iii) Divide by $\frac{1}{4}\sum_c R_c = 2^{b-2}$. ☐

The identity in (ii) is worth emphasising: the spread is not merely bounded below, it is *determined*, and only by the parity of b — the residues of the individual elements are irrelevant. We have checked it numerically for the truncated prime sequences and for random odd sequences up to $k = 32$, where the ratio of the measured relative spread to $2\sqrt{2} \cdot 2^{-b/2}$ is exactly 1 when b is odd and exactly $\sqrt{2}$ when b is even, with no intermediate values.

Two remarks. First, the vanishing $F_B(-1) = 0$ used in the proof is Theorem 4.2 (no flat edges) in Fourier form, so the parity structure that makes lm well defined is the same structure that forces the modulus-4 obstruction.

Second, the contrast with the modulus 6 is the point. The factor $1 + \zeta_6^a$ *can* vanish, namely when $a \equiv 3 \pmod 6$, and then the whole modulus-6 mode disappears. The factor $1 + i^a$ never vanishes for odd a: the modulus-4 obstruction is present in every odd sequence, and by (ii) its size is the same for all of them.

What Theorem 7.6 does and does not say

We must be careful about which quantity is being bounded, and we state the limitation explicitly because it is easy to overread the theorem.

Theorem 7.6 concerns the *aggregate* counts $R_c = \sum_{m\equiv c\,(4)} r_B(m)$, sums over full residue classes. The quantity appearing in Theorem 7.3 is different: ε_d is a *pointwise* oscillation over the bounded window Win_d. The former is a statement about the global distribution of r_B modulo 4; the latter about its behaviour on $O(1)$ many points. Neither implies the other formally, and we do not claim a lower bound for ε_d.

What we can say is that the two live at the same scale, and that the numerics bear this out. Write $L(b) = 2\sqrt{2} \cdot 2^{-b/2}$ for the quantity in (iii). For random odd sequences the measured ε_2 satisfies $\varepsilon_2/L(b) \in [0.85, 1.77]$ over $k = 16, \ldots, 32$, hovering about 1 — and note that it dips *below* 1 at $k = 24$, which is possible precisely because L bounds the aggregate spread and not ε_2. For the primes, by contrast, $\varepsilon_2/L(b)$ grows from 11.7 at $k = 16$ to 293.8 at

$k = 32$, since the modulus-6 obstruction decays at rate $\sqrt{3}/2 \approx 0.866$ against the modulus-4 rate $1/\sqrt{2} \approx 0.707$.

So the informal reading is: random odd sequences are about as flat as the modulus-4 obstruction permits, and the primes are two orders of magnitude less flat than that, for the arithmetic reason isolated in Proposition 10.5. One cannot, however, upgrade the first half of that sentence to a theorem covering all odd sequences: no window analogue of Theorem 7.6 holds, and Remark 10.2 exhibits sequences with $\varepsilon_2 = 0$ exactly.

7.4 Flatness measured

We computed ε_d exactly, by dynamic programming over $\mathbb{Z}$, for the initial segments of the odd primes and for random odd sequences drawn from the same range, at $k = 8, 10, \ldots, 24$ and $n = \lfloor T/2 \rfloor$. Table 1 gives the primes.

k	deg	lm	lm / deg	W_D	ε_1	ε_2	ε_3	ε_4
14	126	628	4.9841	5.3486	0.0328	0.3125	0.3200	0.5833
16	394	2138	5.4264	5.3492	0.0051	0.1827	0.2386	0.6944
18	1290	6706	5.1984	5.3493	0.0031	0.1518	0.1944	0.4122
20	4344	22808	5.2505	5.3493	0.0007	0.1104	0.1462	0.2987
22	14846	80008	5.3892	5.3493	0.0009	0.0734	0.1112	0.2283
24	51068	270392	5.2947	5.3493	0.0003	0.0570	0.0789	0.1573

Table 1: Exact flatness ε_d for $A =$ first k odd primes, $n = \lfloor T/2 \rfloor$.

The flatness decays geometrically; see Figure 1. Log-linear regression on $k \geq 14$ gives the decay rates per unit increase of k shown in Table 2.

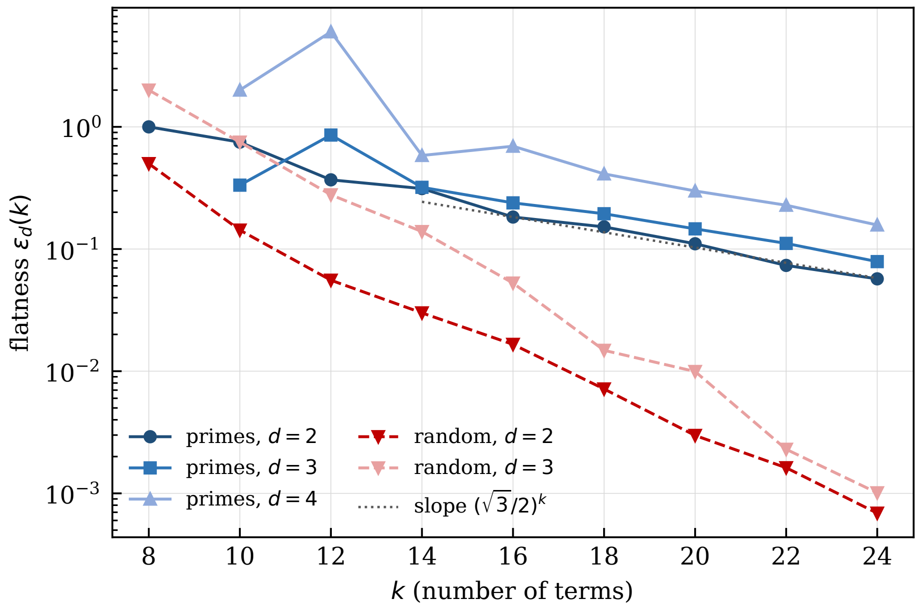

Figure 1: Exact flatness $\varepsilon_d(k)$ on a logarithmic scale. The primes (blue) track the reference slope $(\sqrt{3}/2)^k$ predicted by the modulo-6 heuristic of §7.4(i); random odd sequences (red, median over 20 seeds) decay strictly faster.

Two features stand out.

d	$\lambda(\varepsilon_d)$, primes	R^2	λ(ripple)	R^2	$\lambda(\varepsilon_d)$, random
2	0.8478	0.986	0.8590	0.989	0.683
3	0.8721	0.996	0.8577	0.988	0.611
4	0.8643	0.929	0.8569	0.990	—
6	—	—	0.8684	0.983	—

Table 2: Decay rate λ of $\varepsilon_d(k)$ and of the modulo-6 ripple amplitude, from log-linear regression on $k \geq 14$. Compare $\sqrt{3}/2 = 0.86603$.

(i) The rate matches $\sqrt{3}/2$. We have $\sqrt{3}/2 = 0.86603$, and the measured ripple rates lie in $[0.857, 0.868]$, i.e. within 1.1%. Here is the heuristic that predicts it. Every prime ≥ 5 is $\equiv \pm 1 \pmod 6$. Consequently the generating polynomial of B_d evaluated at a primitive sixth root of unity ζ is $\prod_{a \in B_d}(1 + \zeta^a)$ with each factor of modulus $|1 + \zeta^{\pm 1}| = \sqrt{3}$, against 2 for the trivial character. The non-trivial character therefore contributes a ripple of relative amplitude $(\sqrt{3}/2)^{|B_d|}$ to r_{B_d}, periodic with period 6. Since $|B_d| = k - N(d)$ and $N(d)$ is bounded for fixed d, the amplitude decays like $(\sqrt{3}/2)^k$. The measurement confirms both the rate and the period.

(ii) Random sequences are flatter. Random odd sequences show $\lambda \approx 0.68$ and 0.61, i.e. strictly faster decay, and at $k = 24$ their median ε_2 is 0.0007 against 0.0570 for the primes — a factor of 80. The primes are subject to an arithmetic obstruction that generic sequences are not.

This is the sense in which the primes are simultaneously *smoother* and *less flat* than random sequences: smoother at first order, because Γ is small (§8), but less flat at second order, because of the modulo-6 bias. We isolate the resulting conjecture in §10.

8 Numerical experiments

8.1 Setup

All landscapes below are enumerated exhaustively. For $k \leq 20$ we used a compiled Lean executable enumerating all 2^k configurations (the full 100-seed sweep over $k \in \{8, 12, 16, 18, 20\}$ takes 11.5 s); for $k \leq 24$ we used exact dynamic programming over $\mathbb{Z}$ via Theorem 6.3, cross-validated against the enumeration at $k = 20$ (both give $\mathrm{lm} = 22808$, $\deg = 4344$). Random sequences are drawn without replacement from the odd integers in $[3, a_k]$, matching the range of the primes of the same length. The target is $n = \lfloor T/2 \rfloor$ throughout.

8.2 lm / deg converges to Γ

Put $Q = (\mathrm{lm} / \deg)/\Gamma$. Theorem 7.3 predicts $Q \in [(1+\varepsilon)^{-1}, 1+\varepsilon]$ up to the negligible tail term. Table 3 and Figure 2 record the behaviour.

k	mean Q	s.d.	slope a (s.e.)	intercept b	R^2	corr(Γ, lm)
8	1.112	0.498	1.907 (0.275)	−4.757	0.329	0.843
12	1.002	0.080	1.050 (0.038)	−0.342	0.884	0.956
16	0.993	0.025	1.033 (0.014)	−0.313	0.982	0.951
18	0.992	0.014	1.028 (0.007)	−0.281	0.995	0.982
20	**0.995**	**0.0067**	**1.013 (0.003)**	−0.142	**0.99903**	0.981

Table 3: Q over 100 random odd sequences per k; regression of lm / deg on Γ.

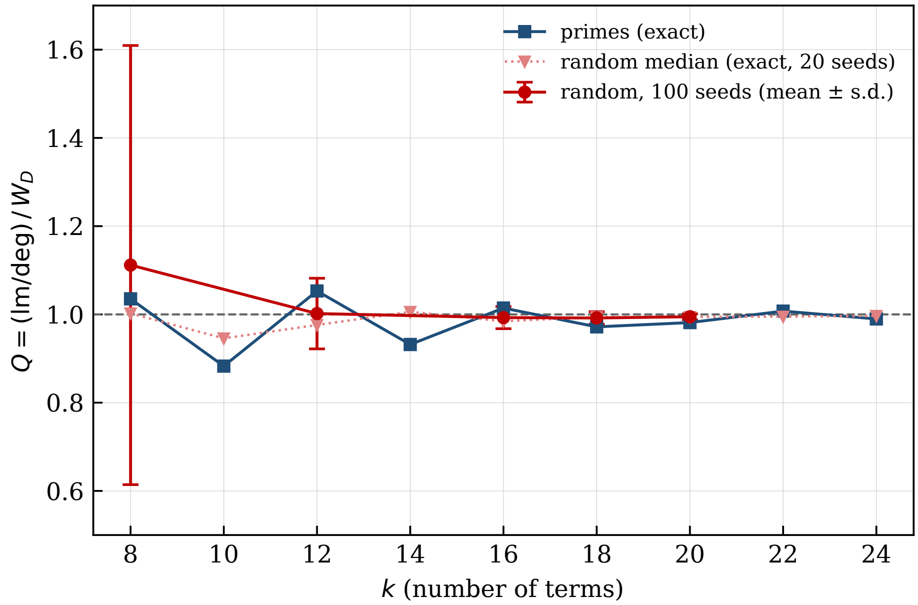

Figure 2: Convergence of $Q = (\mathrm{lm} / \deg)/W_D$. Error bars are one standard deviation over 100 random sequences; they contract steadily, consistent with $\varepsilon \to 0$. The primes (blue) oscillate about 1 rather than converging monotonically; see §8.4 and Conjecture 10.6.

The dispersion contracts steadily. Correlation between Γ and $\mathrm{lm} / \deg$ reaches 0.9995 at $k = 20$.

8.3 Primes against random

k	8	12	16	18	20
$z(\mathrm{lm})$	-1.43	-0.96	-1.17	-1.09	-1.14
percentile of lm	10	21	8	5	4
$z(\Gamma)$	-0.76	-1.20	-1.30	-1.09	-1.09

Table 4: Position of the primes within the 100 random instances.

The primes have fewer valleys than a random sequence of the same length and range at every k tested, but the effect is of size ≈ 1 standard deviation and is not individually significant. What Table 4 does show clearly is that $z(\mathrm{lm}) \approx z(\Gamma)$: whatever smoothness the primes possess is accounted for by their gap series, exactly as Theorems 5.1 and 7.3 predict.

8.4 An oscillation

The exact computation at $k \leq 24$ reveals a feature invisible at the resolution of Table 3. For the primes,

$$Q(k) = 1.035,\ 0.883,\ 1.053,\ 0.932,\ 1.014,\ 0.972,\ 0.982,\ 1.007,\ 0.990 \qquad (k = 8, 10, \ldots, 24),$$

oscillating about 1 rather than converging monotonically, whereas the median Q over random instances increases monotonically from below to 0.9967 at $k = 24$. We believe this oscillation is the signature of the modulo-6 ripple of §7.4 beating against the target $n = \lfloor T/2 \rfloor$; see Conjecture 10.6.

9 Comparison with the coefficient of variation

Alyahya and Rowe [AR14] reported that the number of local optima in the 1-bit-flip NPP landscape correlates strongly and negatively with the coefficient of variation $CV = \mathrm{sd}(a)/\mathrm{mean}(a)$ of the weights, and proposed an estimator on that basis. Using the 100-seed data of §8 we first reproduce their finding, then separate the two predictors.

k	$\mathrm{corr}(CV, \mathrm{lm})$	$\mathrm{corr}(\Gamma, \mathrm{lm})$	$\mathrm{corr}(CV, \Gamma)$	$\mathrm{partial}(\Gamma, \mathrm{lm} \mid CV)$	$\mathrm{partial}(CV, \mathrm{lm} \mid \Gamma)$
8	-0.874	$+0.843$	-0.927	$+0.178$	-0.461
12	-0.878	$+0.956$	-0.907	$+0.790$	-0.092
16	-0.757	$+0.951$	-0.836	$+0.887$	$+0.225$
18	-0.741	$+0.982$	-0.807	$+0.967$	$+0.451$
20	-0.793	$+0.981$	-0.838	$+0.952$	$+0.279$

Table 5: Correlations and partial correlations, 100 random instances per k.

For $\mathrm{lm} / \deg$ — the quantity our theorems actually control — the separation is sharper still: at $k = 20$, $\mathrm{partial}(\Gamma, \mathrm{lm} / \deg \mid CV) = +0.9986$ ($p = 2.4 \times 10^{-126}$) while $\mathrm{partial}(CV, \mathrm{lm} / \deg \mid \Gamma) = +0.306$.

For $k \geq 12$, conditioning on CV leaves the predictive power of Γ essentially intact, whereas conditioning on Γ removes most of that of CV. We read this as: CV is a proxy for Γ, useful because the two are strongly correlated (≈ -0.85) for random sequences, but not the operative quantity.

This is what Theorem 5.1 predicts. CV is a function of the multiset of weights; Γ is not. The window measure W_D, being a function of the multiset, agrees with Γ *only for the increasing ordering* (Example 5.2). Any order-blind statistic must therefore fail to equal W_D in general, and CV can capture Γ only to the extent that, for the random model in question, order-blind data happen to determine the increasing rearrangement's gap series in distribution.

A concrete illustration: for the primes, $z(CV)$ is $+0.85, +1.52, +1.70, +1.51, +1.57$ at $k = 8, \ldots, 20$ — *positive*, i.e. the primes are more dispersed than random. Both accounts therefore predict "primes are smooth", but only Γ predicts the magnitude, and only Γ appears in an identity.

10 Open problems

Problem 10.1 (Asymptotic flatness)**.** Show that for the sequences of interest the flatness ε_k of Definition 7.2 tends to 0, so that Theorem 7.3 yields $\mathrm{lm}_A(n)/\deg_A(n) \to \Gamma(A)$.

For random odd sequences this appears to be within reach of local central limit theorems together with Erdős–Littlewood–Offord anticoncentration [Er45], which bounds $\max_m r_B(m)$ from above and is exactly the ingredient needed to control the tail hypothesis. For the primes the situation is different. The classical asymptotic for partitions into distinct primes, due to Roth and Szekeres [RS54], gives $\log p(n, \mathcal{P}) \sim \kappa\sqrt{2n/\log n}$; *this is a statement on the logarithmic scale and does not suffice*, since flatness is a statement about *ratios* of r at nearby arguments. What is required is either an asymptotic with the pre-exponential factor for distinct-prime partitions of truncated prime sequences, or a direct exponential-sum estimate of Vinogradov type for r_{B_d}.

Remark 10.2 (There is no window analogue of Theorem 7.6)*.* It is natural to ask whether Theorem 7.6 transfers from the aggregate counts R_c to the bounded windows Win_d, giving a lower bound $\varepsilon_d \gg 2^{-|B_d|/2}$ valid for every odd sequence. It does not: the window flatness can vanish identically. For $d = 2$ and a sequence A with $\min A > 4$ one has $I_2 = \varnothing$, so $\mathrm{Win}_2 = \{n-2, n, n+2\}$ has only three points, and r_{B_2} can be made constant on them. For example, with

$$A = (7, 15, 21, 45, 49, 55, 67, 69, 73, 75, 81, 85, 89, 93, 97, 105, 107, 113, 115, 131, 135, 141, 145, 147, 151, 155, 159, 1$$

($k = 28$) all three values equal 377736, so $\varepsilon_2 = 0$ exactly. The agreement is to one part in 3.8×10^5 and is therefore not an artefact of the integrality of r; sequences with $\varepsilon_2 = 0$ were found by local search at every size tested, with the common value growing from 10 at $k = 12$ to 377736 at $k = 28$.

The reason is simply that Win_d imposes $O(1)$ constraints on a sequence with k degrees of freedom. Theorem 7.6 constrains the *global* distribution of r_B modulo 4 and says nothing about any fixed finite set of points. What survives is the statement for the sequences we care about: for random odd sequences ε_2 sits at the modulus-4 scale (§7.3), and for the primes it sits two orders of magnitude above it — but neither is forced by Theorem 7.6 alone.

Conjecture 10.3 (Ripple amplitude)**.** For the odd primes and fixed $d \geq 2$,

$$\frac{r_{B_d}(m)}{\mathrm{Main}(m)} - 1 \;=\; 2\Big(\tfrac{\sqrt{3}}{2}\Big)^{|B_d|+1} \cos\Big(\tfrac{\pi m}{3} - \tfrac{\pi}{6}(c_1 - c_5)\Big) + o\big((\sqrt{3}/2)^k\big),$$

where Main is the local-CLT Gaussian and $c_j = \#\{a \in B_d : a \equiv j \bmod 6\}$; consequently $\varepsilon_d(k) \asymp (\sqrt{3}/2)^k$.

The arithmetic input here is not conjectural. Write $\zeta = e^{2\pi i/6}$ and $F_B(z) = \prod_{a \in B}(1 + z^a)$, so the r_B are the coefficients of F_B.

Proposition 10.4 (Character values)**.** $|1 + \zeta^a|$ *depends only on* $a \bmod 6$*, taking the values* $2, \sqrt{3}, 1, 0, 1, \sqrt{3}$ *for* $a \equiv 0, \ldots, 5$*. Hence if every element of* B *is a prime* ≥ 5 *then* $F_B(\zeta) = 3^{|B|/2} e^{i(\pi/6)(c_1 - c_5)}$*, so that* $|F_B(\zeta)|/2^{|B|} = (\sqrt{3}/2)^{|B|}$ exactly*; if* $3 \in B$ *then* $F_B(\zeta) = 0$*; and* $F_B(-1) = 0$ *for any odd* B*, which is Theorem 4.2 in Fourier form.*

Proof. $|1 + e^{i\pi a/3}|^2 = 2 + 2\cos(\pi a/3)$ takes the stated values. Every prime ≥ 5 is $\equiv \pm 1 \pmod 6$ and $1 + \zeta^{\pm 1} = \sqrt{3}\, e^{\pm i\pi/6}$; multiplying gives the formula. For $a \equiv 3$ the factor $1 + \zeta^3$ vanishes, and $\zeta^3 = -1$ gives the last claim. □

What remains conjectural is only the transfer from this Fourier datum to pointwise ratios. The exponent $|B_d| + 1$ rather than $|B_d|$ comes from the width factor $\sqrt{V_0/V_6} = \sqrt{3}/2$ at the sub-arc, where $V_0 = \frac{1}{4}\sum a^2$ and $V_6 = \frac{1}{4}\sum a^2 \sec^2(\pi a/6) = \frac{1}{3}\sum a^2$. We have tested both parts numerically for $k \leq 32$ by extracting the period-6 Fourier component of $r_{B_d}/\mathrm{Main} - 1$.

The measured phase agrees with $(\pi/6)(c_1 - c_5)$ to four decimal places, including at $k = 16$, where $c_1 - c_5 = -3$ moves the predicted phase to $-\pi/2$.

For the amplitude, the extraction is sensitive to the width of the window over which the Fourier component is taken, because the Gaussian reference degrades in the tails: at $k = 26$ the ratio to $2(\sqrt{3}/2)^{|B_d|+1}$ is 0.972, 0.991, 1.056 for half-widths 24, 90, 180 respectively, while $\sigma = \sqrt{V_0} \approx 147$. Taking a narrow window (half-width 24, about 0.15σ, still eight full periods) the ratio is stable in the window and approaches 1; the corresponding ratios to $2(\sqrt{3}/2)^{|B_d|}$ move *away* from 1, which is what identifies the extra factor $\sqrt{3}/2$.

The approach to 1 splits into two branches according to the value of $c_1 - c_5$, which is also the phase:

$c_1 - c_5 = -1:$	0.965, 0.968, 0.970, 0.972, 0.974, 0.976, 0.981, 0.981	$(k = 20, \ldots, 30, 36, 38)$	from below,
$c_1 - c_5 = -3:$	1.025, 1.019, 1.015	$(k = 32, 34, 40)$	from above.

Both branches converge to 1, from opposite sides. Since the phase itself is reproduced to within 7×10^{-4} in both branches, the residual is not an error in the phase but a *phase-coupled* correction to the amplitude — which is what one expects from interference between the leading sub-arc term and the next one. We regard this as evidence for Conjecture 10.3 and as a constraint on any refinement of it.

We have also verified the width factor at a second modulus. For $q = 4$ the prediction is $\sqrt{V_0/V_4} = 1/\sqrt{2}$, and measuring the period-4 component on the stratum $d = 1$ of the primes (where $3 \in B_1$ kills the period-6 mode) gives ratios 0.91–0.97 to the prediction *with* the width factor against 0.64–0.69 *without* it; for random odd sequences the corresponding medians are ≈ 1.05 and ≈ 0.75. The modulus $q = 3$ cannot be tested this way: for any odd sequence coprime to 3 one has $\rho_3 = (1/2)^{|B|}$, which is dominated by $\rho_4 = (1/\sqrt{2})^{|B|}$, so the period-3 component never rises above the period-4 one.

Proposition 10.5 (Extremality of the modulus 6)**.** *Let $M(q)$ denote the geometric mean of $|\cos(\pi r/q)|$ over the r coprime to q. Then*

$$M(q) = \tfrac{1}{2}\, |\Phi_q(-1)|^{1/\varphi(q)},$$

where Φ_q is the q-th cyclotomic polynomial, and

$$\sup_{q \geq 3} M(q) = \frac{\sqrt{3}}{2},$$

attained at $q = 6$ and nowhere else. The next largest values are $M(10) = 5^{1/4}/2$, $M(4) = 1/\sqrt{2}$ and $M(14) = 7^{1/6}/2$. (Lean, finite part: `M12_six_maximal`, `M12_gap`.)

Proof. Since $\prod_{r \perp q}(x - \zeta_q^r) = \Phi_q(x)$, evaluating at $x = -1$ and using $|1 + \zeta_q^r| = 2|\cos(\pi r/q)|$ gives $|\Phi_q(-1)| = 2^{\varphi(q)} \prod_{r \perp q} |\cos(\pi r/q)|$, which is the identity. For the supremum use the classical values: $\Phi_q(-1) = p$ when $q = 2p^j$ with p an odd prime, $\Phi_q(-1) = 2$ when $q = 2^j$ with $j \geq 2$, and $\Phi_q(-1) = 1$ for all other $q \geq 3$. Hence $M(q) = 1/2$ except in two families. For $q = 2p^j$ we get $M(q) = \frac{1}{2} p^{1/(p^{j-1}(p-1))}$, which is largest when the exponent is largest, i.e. at $j = 1$; and $p \mapsto p^{1/(p-1)}$ decreases on odd primes ($3^{1/2} > 5^{1/4} > 7^{1/6} > \cdots$), so the family is maximised at $q = 6$ with value $\sqrt{3}/2$, the runner-up being $q = 10$. For $q = 2^j$, $j \geq 2$, we get $M(q) = \frac{1}{2} \cdot 2^{1/2^{j-1}}$, maximised at $j = 2$ with value $1/\sqrt{2} < \sqrt{3}/2$. □

We verified the identity numerically for $3 \leq q \leq 40$ to machine precision, obtaining $M(6) = 0.866025$, $M(10) = 0.747674$, $M(4) = 0.707107$, $M(14) = 0.691544$. The finite comparison is also formalised: raising to the twelfth power clears denominators, giving $729, 125, 64, 49, 8$ over 4096 for $q = 6, 10, 4, 14, 8$ and $1/4096$ otherwise, so the maximality of $q = 6$ reduces to a comparison of rationals.

The number-theoretic content is this. For a sequence equidistributed among the residues coprime to q, $M(q)$ is the decay rate of $|F_B(\zeta_q)|/2^{|B|}$; Proposition 10.5 then says that $q = 6$ is the slowest mode available to any such sequence. The primes are confined to exactly two residues modulo 6, both of which realise the maximal $|\cos|$, and no other modulus offers a comparable concentration — which is why the bias modulo 6, and no other congruence obstruction, governs the flatness of the primes. The step from equidistribution to the decay rate is the one genuinely analytic ingredient here, and we do not prove it; see Problem 10.1.

Conjecture 10.6 (Oscillation)**.** The oscillation of $Q(k)$ for the primes observed in §8.4 is caused by the ripple of Conjecture 10.3 beating against $n = \lfloor T/2 \rfloor$. Precisely, writing A_d, φ_d for the amplitude and phase in Conjecture 10.3 and $P_d = \prod_{a \in I_d}(1 + e^{-i\pi a/3})$,

$$\frac{\mathrm{lm}_A(n)}{\deg_A(n)} - W_D(A) \;\approx\; \sum_{d \geq 1} 2^{-N(d)}\, \mathrm{Re}\Big[A_d e^{-i\varphi_d}\big(e^{i\pi(n+d)/3} + e^{i\pi(n-d-\sigma_d)/3} - 2e^{i\pi n/3} P_d 2^{-N(d)}\big)\Big].$$

Evaluating the right-hand side for $k \leq 26$ reproduces the sign of the observed deviation in 9 of the 10 cases (all but $k = 8$) and its magnitude to within a factor 0.5–1.7, with a systematic under-estimate of about 25% that we attribute to higher-order sub-arc corrections. The stratum $d = 2$ dominates, and $d = 1$ contributes exactly zero because $3 \in B_1$ forces $F_{B_1}(\zeta) = 0$ by Proposition 10.4 — which also explains why ε_1 is two orders of magnitude below ε_2 in Table 1.

We warn against the simpler guess that the sign is governed by T mod 6 alone: for the primes $T \equiv 2 \pmod 6$ throughout $k = 8, \ldots, 14$ and $k = 18, \ldots, 26$, yet the sign of the deviation alternates across that range. The dependence on $c_1 - c_5$ and on σ_d in the displayed formula is essential.

Problem 10.7 (Asymptotic smoothness of the primes)**.** Determine whether $\operatorname{lm}_{P_k}(n)/\operatorname{lm}_{R_k}(n) \to 0$ as $k \to \infty$, where P_k is the first k odd primes and R_k a random odd sequence of the same length and range. By Theorem 7.3 this reduces to comparing $\Gamma(P_k)$, which converges (to ≈ 5.3493), with $\mathbb{E}[\Gamma(R_k)]$, which we expect to grow like twice the mean gap, i.e. $\asymp \log k$.

Problem 10.8 (Off-centre targets)**.** For $n/T = \rho \neq 1/2$ the factor $2^{-N(d)}$ in the window series is replaced by $(1-\rho)^{N(d)}$, deforming Γ into the family $\Gamma_x(A) = \sum_j a_j x^j$ evaluated at $x = 1 - \rho$. Develop the corresponding theory, including the large-deviation rate function that governs $\operatorname{lm}/2^k$ away from the centre.

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A The Lean development

The formalisation targets Lean 4 (toolchain `leanprover/lean4:v4.32.2`) with Mathlib pinned to the matching tag. Building with `lake build` verifies every declaration below. Each is checked with `#print axioms` to depend only on `propext`, `Classical.choice` and `Quot.sound`; no declaration depends on `sorryAx`.

Paper	Lean declaration	File
Prop. 2.1	`mem_subsetSums_ofFn_iff`	`Theory/Bridge.lean`
Prop. 2.2	`gs_ofFn_eq_energy_min`	`Theory/Bridge.lean`
Prop. 2.3	`gs_eq_zero_iff`	`Theory/Symmetry.lean`
Thm. 3.1	`isStrictLocalMin_iff_of_ge` etc.	`Theory/Landscape.lean`
Cor. 3.2	`isStrictLocalMin_of_exact`	`Theory/Landscape.lean`
Cor. 3.3	`lm_eq_lmClassCount`	`Theory/Landscape.lean`
Thm. 4.1	`gs_compl`; `lm_compl`	`Theory/Landscape.lean`; `Theory/Symmetry.`
Thm. 4.2	`no_flat_edge`	`Theory/Symmetry.lean`
Cor. 4.3	`isStrictLocalMin_iff_le_of_odd`	`Theory/Symmetry.lean`
Thm. 5.1	`windowSeries_eq_gapSeries`	`Theory/Bridge.lean`
Lem. 6.1	`fiber_over`, `fiber_under`	`Theory/Decomposition.lean`
Lem. 6.2	`overCount_eq_zero` etc.	`Theory/Decomposition.lean`
Thm. 6.3	`lm_eq_sum_strata`, `lm_eq_lmPartial`	`Theory/Decomposition.lean`, `Theory/Total`
Thm. 6.4	`fiber_card_eq_repCount`	`Theory/Fiber.lean`
Cor. 6.5	`degCount_eq_sum_repCount`	`Theory/Fiber.lean`
Cor. 6.6	`overCount_eq_repCount` etc.	`Theory/Sandwich.lean`
Lem. 7.1	`ratio_bounds_of_flat`	`Theory/Sandwich.lean`
Thm. 7.3	`sandwich_partial`, `sandwich_total`	`Theory/Sandwich.lean`, `Theory/Total.lean`
Thm. 7.6(i)	`normSq_one_add_I_pow_odd`	`Theory/Ripple.lean`
Thm. 7.6(iii)	`max_abs_cos_sin_sq_ge`, `spread_lower_bound`	`Theory/Ripple.lean`
Prop. 10.5 (finite)	`M12_six_maximal`, `M12_gap`	`Theory/Ripple.lean`
Prop. 10.4	`normSq_one_add_exp` and specialisations	`Theory/Ripple.lean`

The development also contains machine-checked numerical cross-checks (`#eval`), among them: the identity of Theorem 6.3 for $A = (3, 5, 7, 11, 13, 17)$ at every $n \le 56$; the coincidence of the strict and weak local-minimum counts for the same A (Corollary 4.3); the window identity for the first twenty odd primes, both sides evaluating to 1402281/262144; and the failure of that identity for the reordering $(5, 3)$ of Example 5.2.